

FINAL CYBERSECURITY INTERNSHIP REPORT (WEEK 1–3)

Cybersecurity Internship Final Report

User Management System Security Analysis & Implementation

Name: Talha

Role: Cybersecurity Intern

Project: Secure Web Application (User Management System)

Duration: Week 1 – Week 3

Year: 2026

1. PROJECT OVERVIEW

This project focuses on analyzing, securing, and testing a User Management System using cybersecurity principles. The objective was to identify vulnerabilities, implement security measures, and perform final security validation.

The system was developed using Node.js, Express, and MongoDB with authentication and authorization features.

2. WEEK 1 - SECURITY ASSESSMENT

Tasks Performed

- Set up a Node.js based User Management System
- Explored application endpoints (signup, login, profile)
- Performed vulnerability testing using Postman and browser tools

Vulnerability Testing

- Cross-Site Scripting (XSS) test using script injection
- Basic SQL injection attempt in login fields
- Observed application behavior under malicious input



Findings

- Application allowed input without strict validation
- No input sanitization initially
- Weak security awareness in form handling



Week 1 Outcome

Identified basic security vulnerabilities and documented areas for improvement.



3. WEEK 2 - SECURITY IMPLEMENTATION



Tasks Performed

- Implemented password hashing using bcrypt
- Added JWT authentication system
- Created protected routes using middleware
- Improved input validation



Security Enhancements

- Passwords securely hashed before database storage
- JWT token generated during login
- Middleware implemented for route protection
- Email and input validation added



Testing

- Verified user registration and login
- Tested JWT token generation
- Accessed protected routes using Authorization header
- Confirmed unauthorized access is blocked



Week 2 Outcome

A secure authentication system was successfully implemented using modern security practices.

4. WEEK 3 - ADVANCED SECURITY & FINAL TESTING

Security Testing

- Tested valid JWT authentication flow
- Verified invalid token rejection (401 error)
- Tested unauthorized access prevention

Logging System

- Implemented Winston logging library
- Stored logs in security.log file
- Tracked system events and errors

Security Checklist

- Input validation implemented
- Password hashing enabled
- JWT authentication secured
- Protected routes verified
- Logging system active

Final Testing Results

- Valid token → Access granted
- Invalid token → Access denied
- No token → Access denied
- System behaves securely under all conditions

Week 3 Outcome

The system was successfully secured and validated using authentication, authorization, and logging mechanisms.

5. FINAL RESULT

The User Management System is now:

- Secure against unauthorized access
- Protected using JWT authentication
- Secured with hashed passwords

- Equipped with logging system
- Tested for vulnerabilities and fixed successfully

6. TECHNOLOGIES USED

- Node.js
- Express.js
- MongoDB
- JWT (jsonwebtoken)
- bcrypt
- Winston
- Postman

7. CONCLUSION

This internship project successfully demonstrated the process of identifying vulnerabilities, implementing security measures, and performing final validation of a web application.

The system is now more secure and follows basic cybersecurity best practices.